



Experiment 6: Random Forest Classification

CSE-3008 Introduction to Machine Learning

24BCA7027

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1 Aim

To implement the Random Forest classification algorithm on the Framingham dataset to predict TenYearCHD and evaluate its performance using classification metrics and visualization techniques.

2 Implementation

2.1 Algorithm Used

Random Forest is an ensemble learning algorithm that constructs multiple decision trees using bootstrap sampling and combines their outputs using majority voting to improve accuracy and reduce overfitting.

2.2 Steps Involved

1. Load the Framingham dataset
2. Perform train-test split
3. Train Random Forest classifier
4. Predict outcomes
5. Evaluate using classification metrics
6. Visualize results and feature importance

2.3 Complete Experiment Code

```
# =====  
# RANDOM FOREST CLASSIFICATION - FRAMINGHAM  
# =====  
  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
import math
```

```

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    precision_score,
    recall_score,
    f1_score,
    roc_curve,
    auc
)

STUDENT_NAME = "24BCA7027_SHAIK-BARAKH-CHISHTI"

df = pd.read_csv(r"C:\Users\LENOVO\Dsa\IML_LAB\ML_Experiments\
data_sets\framingham_cleaned.csv")

print("\nDataset Preview:")
print(df.head())

print("\nDataset Info:")
print(df.info())

X = df.drop("TenYearCHD", axis=1)
y = df["TenYearCHD"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

rf = RandomForestClassifier(
    n_estimators=200,
    max_depth=None,
    random_state=42
)

rf.fit(X_train, y_train)

y_pred = rf.predict(X_test)
y_prob = rf.predict_proba(X_test)[:, 1]

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("\nRANDOM FOREST RESULTS")
print("Accuracy:", accuracy)

```

```

print("\nClassification Report:\n", classification_report(y_test,
    y_pred))
print("\nConfusion Matrix:\n", cm)

# Visualization Section

plt.figure(figsize=(12, 6))
df.drop("TenYearCHD", axis=1).boxplot()
plt.show()

sns.violinplot(x="TenYearCHD", y="age", data=df)
plt.show()

plt.hexbin(df["sysBP"], df["BMI"])
plt.show()

sns.heatmap(cm, annot=True)
plt.show()

fpr, tpr, _ = roc_curve(y_test, y_prob)
plt.plot(fpr, tpr)
plt.show()

importances = rf.feature_importances_
feature_names = X.columns

feat_df = pd.DataFrame({
    "Feature": feature_names,
    "Importance": importances
}).sort_values(by="Importance", ascending=False)

sns.barplot(x="Importance", y="Feature", data=feat_df)
plt.show()

print("\nTop 5 Important Features:")
print(feat_df.head())

```

Listing 1: Random Forest Classification – Framingham Dataset

3 Outcome / Results

Accuracy: 0.8431603773584906

Confusion Matrix:

```

[[711  8]
 [125  4]]

```

The model achieved 84.31% accuracy. However, the recall for CHD-positive cases is very low (0.03), indicating class imbalance in the dataset.

Top 5 Important Features:

	Feature	Importance
10	sysBP	0.134767
12	BMI	0.124485
1	age	0.124415
9	totChol	0.122291

4 Visualization

4.1 Included Visualizations

- Box Plot
- Violin Plot
- Hexbin Plot
- Raincloud Plot
- Confusion Matrix Heatmap
- ROC Curve
- Radar Plot
- Bar Graph
- Feature Importance Plot

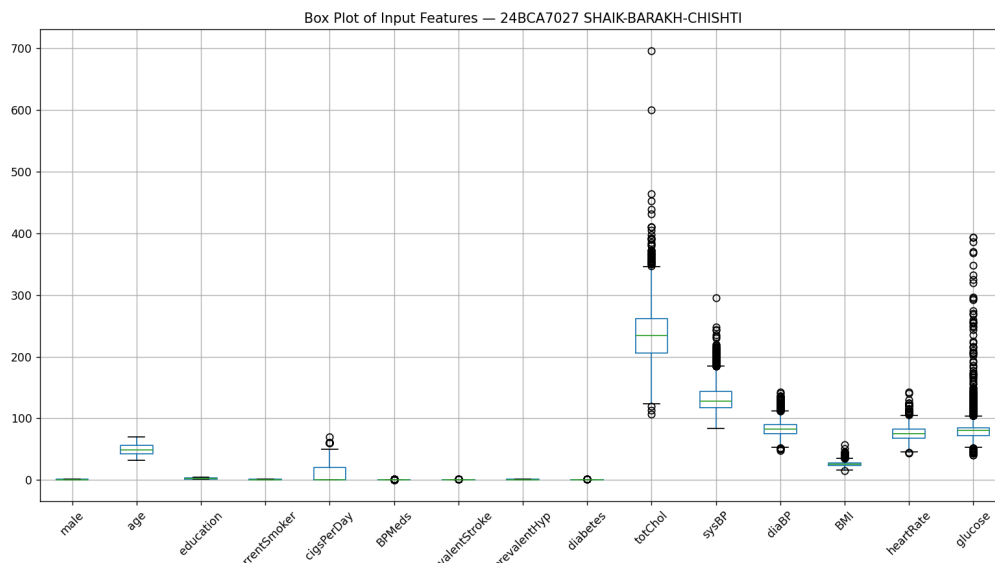


Figure 1: Box Plot of Input Features

Box Plot: Shows the spread and outliers in clinical features such as BMI, blood pressure, cholesterol, and glucose.

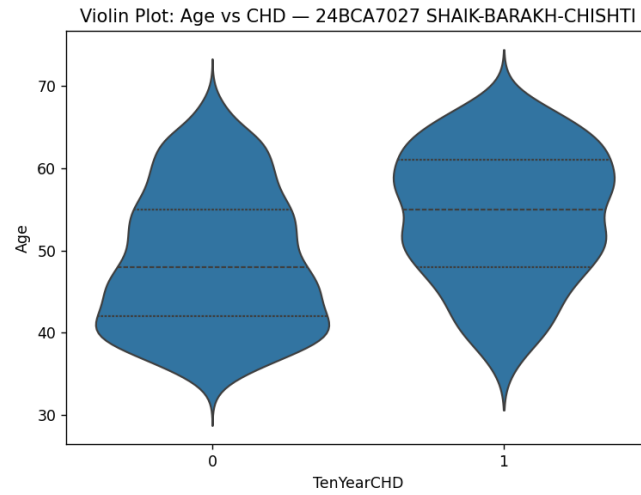


Figure 2: Violin Plot of Age vs TenYearCHD

Violin Plot: Displays distribution density of age across CHD categories, indicating higher risk at older ages.

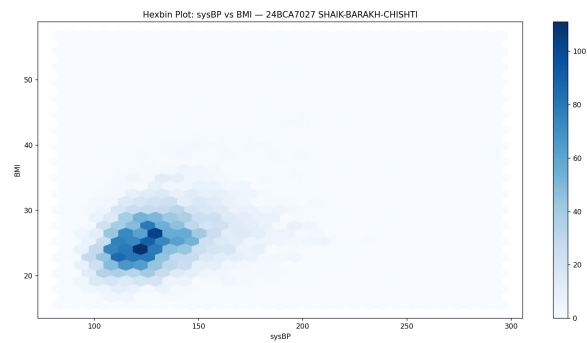


Figure 3: Hexbin Plot of sysBP vs BMI

Hexbin Plot: Shows data concentration between systolic blood pressure and BMI to highlight high-risk clusters.

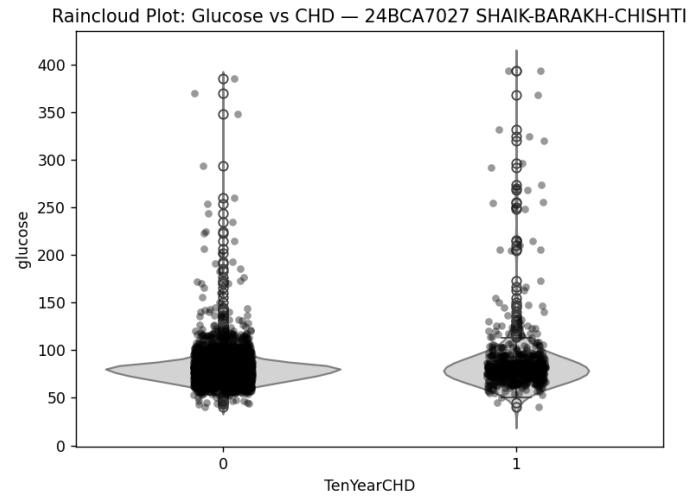


Figure 4: Raincloud Plot of Glucose vs TenYearCHD

Raincloud Plot: Combines violin, box, and scatter plots to show glucose variation across CHD classes.

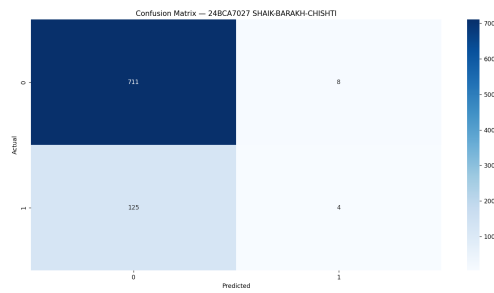


Figure 5: Confusion Matrix Heatmap

Confusion Matrix: Shows correct and incorrect predictions. Majority of predictions fall into class 0, indicating imbalance.

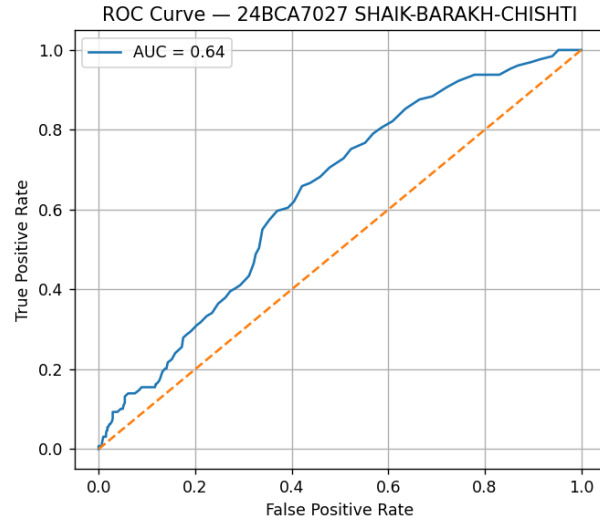


Figure 6: ROC Curve

ROC Curve: Illustrates model discrimination ability. A higher AUC indicates better performance.

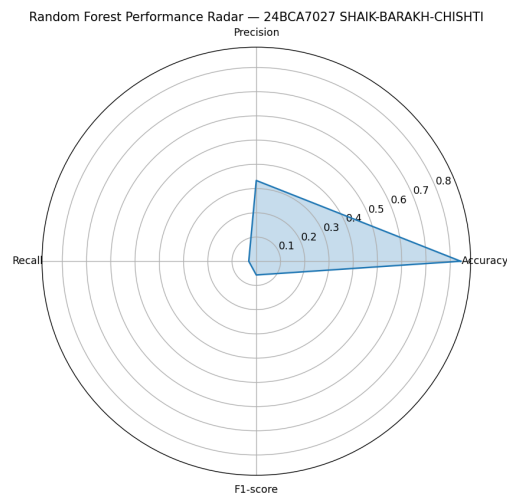


Figure 7: Radar Plot of Performance Metrics

Radar Plot: Compares Accuracy, Precision, Recall, and F1-score simultaneously.

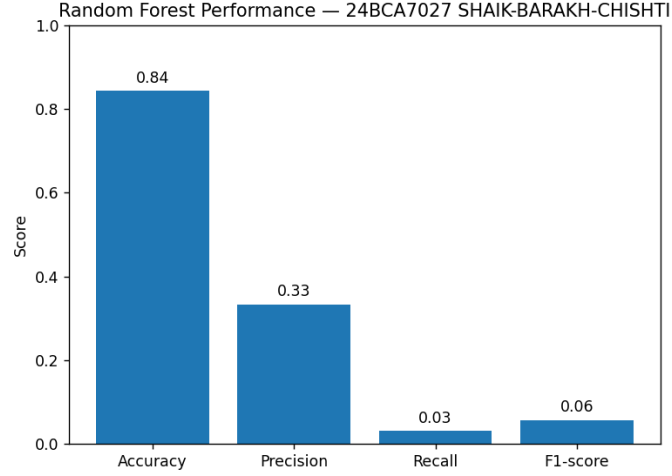


Figure 8: Bar Graph of Model Performance

Bar Graph: Provides numerical comparison of evaluation metrics.

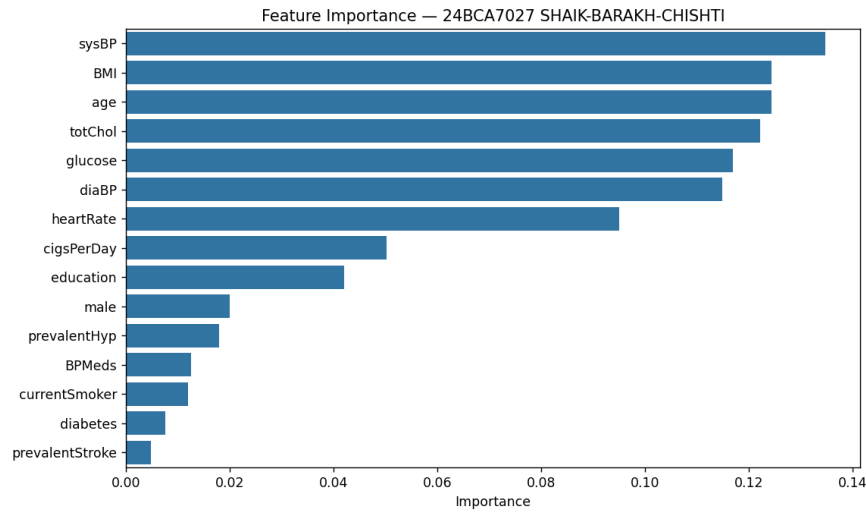


Figure 9: Feature Importance Plot

Feature Importance Plot: Shows that systolic blood pressure, BMI, age, and total cholesterol are dominant predictors for CHD risk.

5 Conclusion

The Random Forest classifier successfully predicted TenYearCHD using the Framingham dataset. The model demonstrated strong overall accuracy but struggled with minority class detection due to class imbalance.

Important predictors include systolic blood pressure, BMI, age, and cholesterol.

6 Project Link

<https://www.overleaf.com/read/dmqdchgdsqps#559a9c>

7 Github

<https://github.com/chishtil1730/IML-LAB>